

Uncertainty and Sentiment in the Filings of ARK ETF Holdings, 2021 to 2026

FRE-GY 7871 A, Assignment 1. 1,805 10-K and 10-Q filings by 93 companies held by the six ARK ETFs on 9 September 2026, scored on the Loughran-McDonald negative and uncertainty word lists. Methods, supplementary tables and further figures are in the separate appendix.

Summary

Negative and uncertain language in the annual reports of ARK holdings rises every year: within company, negative words gain +0.063 pp a year and uncertainty words +0.029 pp (both $p < 0.001$, Table 4). Quarterly reports show no trend in negative words and a small fall in the uncertainty tf.idf score. In quarterly reports that score predicts the following quarter's volatility before the prior-volatility control and weakly after it (+3.1 pp per standard deviation without the control, +2.4 pp with it, $p = 0.043$, Table 5). Negative tone does not predict the four-session return; in quarterly reports the test cannot detect an effect below 3.3 pp per standard deviation (Table 6). Of the quarterly reports whose risk-factor section, Item 1A, can be located, only 47% restate it; 39% refer the reader to the annual report, and that choice moves a word-share measure by itself. Excluding Item 1A cuts the annual-report trends by 27% to 36% and removes the one return association in the data (Section 6). QQQ holdings carry fewer negative and uncertainty words in their annual reports; outside Item 1A the two portfolios write alike (Section 7).

1. Sample

The six ARK portfolios held 121 identifiers on 9 September 2026; the assignment brief counts 124 companies from an earlier snapshot. Removing funds, cash and non-US listings, merging share classes and dropping companies that file 20-F or 40-F leaves 93 SEC filers, 22 of them also held by QQQ, the Invesco Nasdaq-100 ETF. They filed 1,805 original 10-K and 10-Q documents from January 2021 to 9 September 2026, with 0 parse failures; the 53 amendments are never scored. Table 1 lists every filter. The volatility and return samples are filtered separately because a recent filing is scored before its outcome window has elapsed. The assignment's window ends in 2025; the executed notebook and the data workbook hold the same exhibits for that window.

Table 1. Sample filters and the filings each removed

Panel A. From holding identifiers to SEC filers. Absence of 10-K or 10-Q filings is read from EDGAR, never inferred from domicile.

Filter	Removed	Remaining	Unit
Raw holding identifiers	0	121	identifiers
Funds and non-company securities	3	118	identifiers
Foreign local listings	5	113	identifiers
Unresolved SEC identifiers	0	113	identifiers
Merge share classes by company (CIK)	1	112	companies
Files 20-F or 40-F, not 10-K or 10-Q	18	94	companies
No 10-K, 10-Q, 20-F or 40-F on record	1	93	companies

Panel B. From filings to the three analysis samples.

Sample	Filter	Removed	Remaining	Companies
Text	All 10-K and 10-Q filings and amendments	0	1,858	93
	Remove amendments and parse failures	53	1,805	93
	At least 2,000 words (10-K) or 1,000 (10-Q)	0	1,805	93
	One filing per company per quarter, the earliest	9	1,796	93
Volatility	Filings in the text sample	0	1,796	93
	Event day priced and day -1 price at least \$3	126	1,670	93
	Outcome window complete by the market-data cutoff	90	1,580	88
	60 daily returns observed before and after	26	1,554	85

Sample	Filter	Removed	Remaining	Companies
Return	Complete volatility windows before and after	0	1,554	85
	Share count on the filing's cover page	80	1,474	81
	Dollar volume and other controls available	17	1,457	81
	Filings in the text sample	0	1,796	93
	Event day priced and day -1 price at least \$3	126	1,670	93
	Outcome window complete by the market-data cutoff	3	1,667	92
	60 daily returns observed before the filing	30	1,637	89
	Complete four-session return window	0	1,637	89
	Share count on the filing's cover page	87	1,550	85
	Dollar volume and other controls available	17	1,533	85

2. Measures

The lists are the active entries of the March 2026 Loughran-McDonald Master Dictionary: 2,345 negative and 297 uncertainty words (40 on both). The brief's count of 2,355 includes ten negative words since retired. Each filing is scored on each list twice: as a share of its words, and with the tf.idf weighting of equation (1) in Loughran and McDonald (2011), fitted on the ARK corpus, under which a word present in every filing carries no weight. Word shares are in percent and differences in percentage points (pp). ARK annual reports average 2.31% negative and 1.97% uncertainty words (Table 2), against 1.39% and 1.20% for the 1994 to 2008 10-Ks of Loughran and McDonald; the two shares correlate at 0.85 across filings. Table 3 shows why the scorings differ for uncertainty: MAY is 36% of uncertainty counts and appears in 100% of filings, so it counts fully in the word share and not at all in tf.idf.

Table 2. Summary statistics by report type

Word shares in percent; tf.idf in score units. Annual and quarterly reports are kept apart because a 10-K is longer and heavier in risk language.

Form	Measure	N	Mean	SD	P25	Median	P75
10-K	Negative, word share	453	2.310	0.424	2.035	2.317	2.595
10-K	Uncertainty, word share	453	1.973	0.239	1.851	1.994	2.136
10-K	Negative, tf.idf	453	167.616	55.344	121.893	158.812	212.474
10-K	Uncertainty, tf.idf	453	27.240	8.229	22.017	26.195	31.982
10-Q	Negative, word share	1,343	2.031	0.983	1.165	1.739	2.994
10-Q	Uncertainty, word share	1,343	1.820	0.584	1.326	1.647	2.412
10-Q	Negative, tf.idf	1,343	78.975	68.741	21.977	47.704	134.370
10-Q	Uncertainty, tf.idf	1,343	13.560	7.492	7.530	11.581	19.398

Table 3. Thirty most frequent words on each list

Share: the word's count divided by all occurrences of its own list in the ARK text sample.

Rank	Negative word	Share (%)	Uncertainty word	Share (%)	Rank	Negative word	Share (%)	Uncertainty word	Share (%)
1	LOSS	5.06	MAY	35.76	16	NEGATIVE	1.03	DEPEND	0.93
2	ADVERSELY	4.28	COULD	18.48	17	DELAYS	0.98	VOLATILITY	0.91
3	CLAIMS	3.08	RISK	4.85	18	DAMAGES	0.96	UNCERTAINTY	0.90
4	LOSSES	2.95	RISKS	4.47	19	RESTATED	0.92	DIFFER	0.86
5	ADVERSE	2.87	BELIEVE	2.87	20	DELAY	0.91	ANTICIPATE	0.81
6	AGAINST	2.52	APPROXIMATELY	2.52	21	DECLINE	0.89	CONTINGENT	0.80
7	UNABLE	2.02	ASSUMPTIONS	1.90	22	LIMITATIONS	0.86	EXPOSURE	0.74

Rank	Negative word	Share (%)	Uncertainty word	Share (%)	Rank	Negative word	Share (%)	Uncertainty word	Share (%)
8	LITIGATION	2.01	INTANGIBLE	1.49	23	CHALLENGES	0.78	VARIABLE	0.67
9	FAILURE	1.88	POSSIBLE	1.31	24	VOLATILITY	0.77	PENDING	0.66
10	HARM	1.75	UNCERTAINTIES	1.14	25	FINES	0.74	DEPENDS	0.65
11	FAIL	1.37	MIGHT	1.12	26	DISRUPTIONS	0.68	DEPENDENT	0.59
12	PENALTIES	1.18	FLUCTUATIONS	1.07	27	INVESTIGATIONS	0.68	CONTINGENCIES	0.53
13	DIFFICULT	1.18	UNCERTAIN	1.02	28	BREACH	0.67	PROBABLE	0.50
14	NEGATIVELY	1.17	ANTICIPATED	0.99	29	INFRINGEMENT	0.63	VARY	0.48
15	IMPAIRMENT	1.16	PREDICT	0.94	30	HARMED	0.62	ASSUMED	0.48

3. Trends

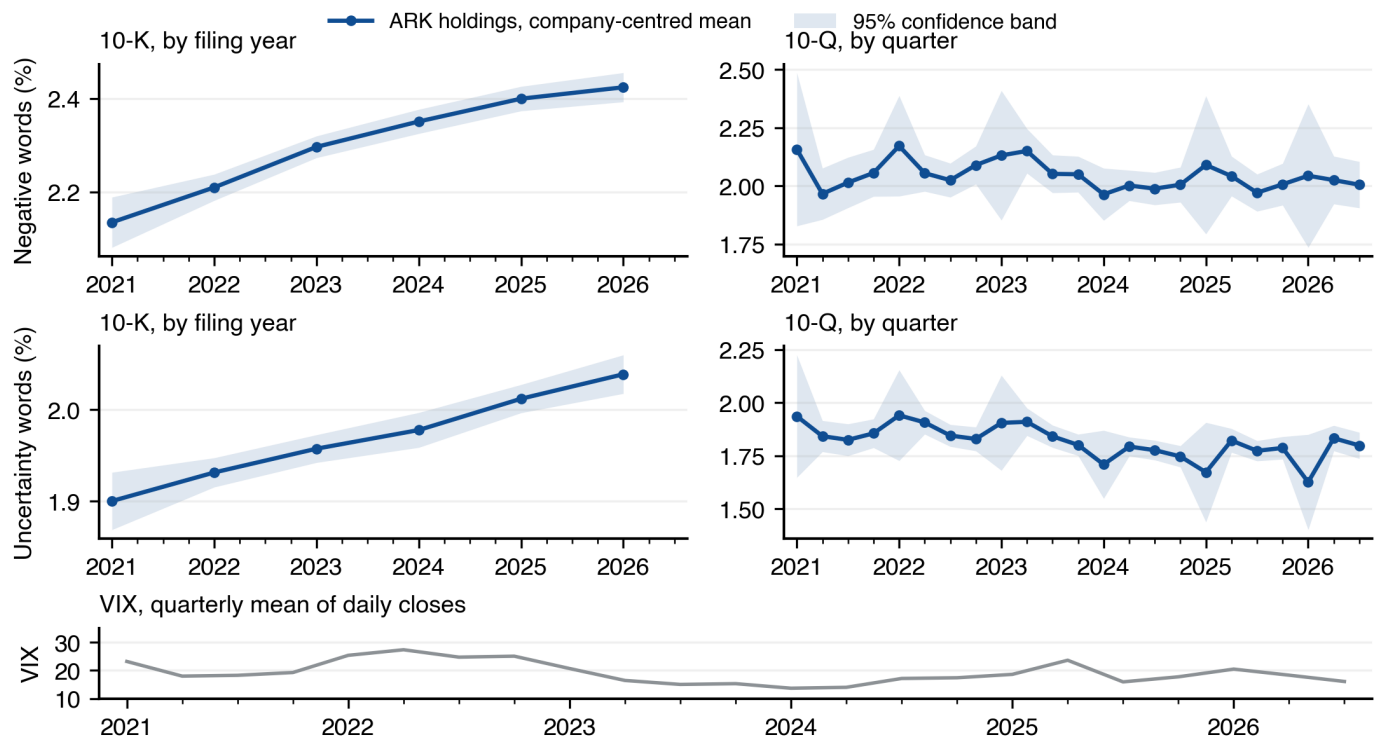


Figure 1. Both measures by quarter, ARK holdings, with the VIX. Lines are company-centred means (each company's score less its own mean, plus the group mean) with 95% bands; annual reports are aggregated by filing year because they cluster in the first calendar quarter. First-quarter 10-Q cells hold 6 to 9 reports, hence their wider bands.

Annual-report tone rises every year: negative words from 2.14% in 2021 filings to 2.42% in 2026, uncertainty words from 1.90% to 2.04%. Quarterly reports show no trend in negative words; their uncertainty share falls on the aggregate test (Newey-West $t = -3.71$) but not within company ($p = 0.112$), the pattern the assignment warns of, and their uncertainty $tf.idf$ score falls on both tests. The VIX, in the bottom strip, peaks in 2022Q2; the report offers no test of its relation to filing tone. Table 4 tests the trends two ways: a regression of the quarterly mean on time with Newey-West (four lags) standard errors, and a within-company regression with company and seasonal effects.

Table 4. Trend tests, aggregate and within company

Slopes per year: percentage points for word shares, score units for tf.idf. Aggregate: regression of the quarterly mean on time, 17 quarters with annual reports and 23 with quarterly reports, Newey-West standard errors. Within company: filing-level, company and seasonal effects, two-way clustered inference.

Report	Measure	Aggregate slope	Newey-West t	Within-company slope	t	p	Filings
10-K	Negative, word share	+0.061	9.68	+0.063	6.71	< 0.001	453
10-K	Uncertainty, word share	+0.024	4.48	+0.029	7.11	< 0.001	453
10-K	Negative, tf.idf	+2.992	2.80	+2.300	3.32	0.004	453
10-K	Uncertainty, tf.idf	-0.231	-1.23	+0.039	0.19	0.849	453
10-Q	Negative, word share	-0.015	-1.77	-0.008	-0.51	0.615	1,343
10-Q	Uncertainty, word share	-0.033	-3.71	-0.017	-1.66	0.112	1,343
10-Q	Negative, tf.idf	-1.243	-1.57	-1.425	-1.09	0.288	1,343
10-Q	Uncertainty, tf.idf	-0.431	-3.13	-0.363	-2.19	0.040	1,343

Believed: the within-company annual-report trends. Both word shares rise at $p < 0.001$ ($t = 6.7$ and 7.1), the aggregate Newey-West test agrees ($t = 9.7$ for negative words against 8.9 with ordinary errors), and the tf.idf score confirms the negative trend ($+2.30$ score units a year, $p = 0.004$). Not believed as a change in how distinctively companies hedge: the uncertainty tf.idf slope is flat ($p = 0.849$), so the rise in uncertainty comes through words such as MAY. Section 6 shows that Item 1A lengthening accounts for 36% of the negative rise and 27% of the uncertainty rise. Quarterly reports show no trend on any scoring except a fall in the uncertainty tf.idf score (-0.36 a year, $p = 0.040$).

4. Uncertainty and post-filing volatility

Table 5. Volatility over the following quarter on uncertainty, with and without prior volatility

Effect: change in annualised volatility, in percentage points, per one standard deviation of the measure. Detectable: the effect this design finds with 80% power at 5%. Controls: log size, log dollar volume, prior excess return; company and calendar-quarter effects.

Report	Measure	Prior volatility	Filings	Effect per SD (pp)	Detectable (pp)	p
All	Uncertainty, word share	No	1,457	+0.35	2.03	0.618
All	Uncertainty, word share	Yes	1,457	+0.09	1.76	0.888
All	Uncertainty, tf.idf	No	1,457	-0.02	3.55	0.988
All	Uncertainty, tf.idf	Yes	1,457	-0.30	3.06	0.778
10-K	Uncertainty, word share	No	386	-4.92	12.77	0.270
10-K	Uncertainty, word share	Yes	386	-6.52	14.29	0.195
10-K	Uncertainty, tf.idf	No	386	-1.86	12.19	0.656
10-K	Uncertainty, tf.idf	Yes	386	-1.44	11.79	0.722
10-Q	Uncertainty, word share	No	1,071	+2.79	3.98	0.053
10-Q	Uncertainty, word share	Yes	1,071	+2.16	3.79	0.111
10-Q	Uncertainty, tf.idf	No	1,071	+3.15	3.33	0.012
10-Q	Uncertainty, tf.idf	Yes	1,071	+2.42	3.28	0.043

The gap between the two estimates is the result. In quarterly reports the tf.idf score adds $+3.15$ pp of annualised volatility per standard deviation without the control ($p = 0.012$) and $+2.42$ pp with it ($p = 0.043$); the word share moves from $+2.79$ pp ($p = 0.053$) to $+2.16$ pp ($p = 0.111$). The control removes about a quarter of each estimate. Annual reports show nothing, with detectable effects of 12 to 14 pp. Believed weakly: 1 of the twelve estimates is significant after the control, at $p = 0.043$, against detectable effects of 3.3 pp; on the quarterly reports whose Item 1A is located, the word-share estimate with the control is $+1.3$ pp ($p = 0.300$) and, excluding the section, $+0.1$ pp ($p = 0.951$).

5. Sentiment and the filing-period return

Table 6. Four-session excess return on negative tone, with controls

Excess return over the S&P 500 ETF from the filing event day to day 3, in percentage points per standard deviation of the measure. Same controls as Table 5 plus prior volatility.

Report	Measure	Filings	Effect per SD (pp)	Detectable (pp)	p
All	Negative, word share	1,533	-0.53	1.63	0.352
All	Negative, tf.idf	1,533	-0.80	1.99	0.253
10-K	Negative, word share	387	-1.64	8.40	0.570
10-K	Negative, tf.idf	387	-3.15	7.01	0.201
10-Q	Negative, word share	1,146	-1.80	3.33	0.129
10-Q	Negative, tf.idf	1,146	-1.57	3.55	0.212

No estimate reaches 5%: the closest is -1.80 pp per standard deviation of the negative word share in quarterly reports ($p = 0.129$). Not believed as evidence of no effect: the test is underpowered, with a detectable effect of 3.3 pp per standard deviation in quarterly reports and 1.6 pp across all filings. Not believed as evidence of an effect either. Section 6 shows that on the quarterly reports whose Item 1A is located the whole-filing estimate is -2.03 pp ($p = 0.024$) and that this is a disclosure effect rather than a tone effect.

6. The risk-factor section

Of 1,211 ARK quarterly reports with a located Item 1A heading (90% of filings), 47% restate, 11% claim no material change but add updates, 39% refer the reader to the annual report and 3% disclose nothing. A restated section runs to a median of 27,137 words, a reference to 79; in annual reports Item 1A is 35% of all words. Item 1A is the most hedged and most negative part of a filing: in restated sections 3.3% of words are uncertainty words and 4.1% negative words, against 1.4% and 1.3% in the rest of the same filings. A word share divides list words by all words, so replacing a 27,000-word section with a 79-word reference removes far more list words than words, and the whole-filing share falls although nothing else in the filing has changed. Across 81 such quarters by 36 companies the uncertainty share falls 0.21 pp and the negative share 0.38 pp, while the share of the rest of the filing moves +0.05 and 0.00 pp. Figure 2 shows the two largest cases; Table 7 ranks the holdings on the measure excluding Item 1A; Table 8 repeats each result on the same filings without Item 1A.

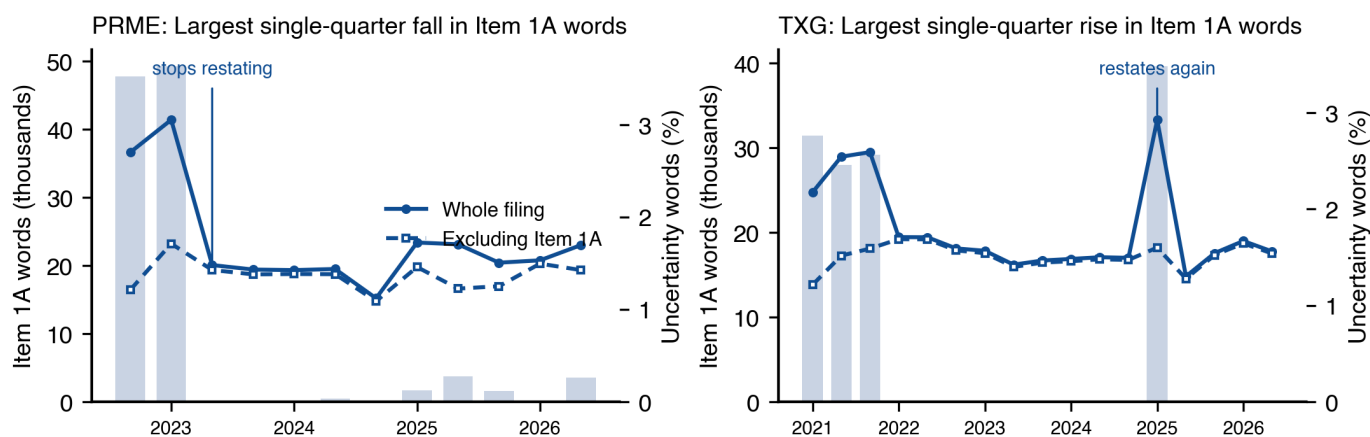


Figure 2. Item 1A words (bars) and uncertainty word share of the whole filing (solid) and excluding Item 1A (dashed, hollow squares), quarterly reports. Left: PRME (Prime Medicine, Inc.), August 2023, 49,292 to 75 words; the whole-filing share falls 1.57 pp, the share excluding Item 1A 0.28 pp. Right: TXG (10X Genomics, Inc.), May 2025, restating after three years, 52 to 39,630 words.

Table 7. ARK holdings ranked on uncertainty words excluding Item 1A

Level: latest annual report with a located Item 1A, 83 companies. Trend: within-company slope over 2021 to 2026 for the 71 companies with at least three such reports; slopes rest on three to six observations and rank companies. Tickers; full lists in Appendix Figure B2 and Table C8.

Rank	Most uncertain (%)	Least uncertain (%)	Largest rise (pp a year)	Largest fall (pp a year)
1	LLY 1.78	IONS 0.85	WGS +0.121	KDK -0.105
2	DKNG 1.76	IRDM 0.94	LUNR +0.110	TER -0.098
3	JOBY 1.61	KDK 1.02	DASH +0.101	META -0.057
4	PACB 1.60	CRWV 1.05	MASS +0.092	TWST -0.032
5	NTRA 1.57	P 1.11	BMNR +0.074	CNTN -0.025

Table 8. The same results on the whole filing and excluding Item 1A, identical filings

Within-company trends and outcome regressions as in Tables 4 to 6, on the filings whose Item 1A heading was located. Word-share measures only.

Result	Filings	Whole filing	p	Excluding Item 1A	p
Annual reports: negative words, trend (pp a year)	404	+0.064	< 0.001	+0.041	< 0.001
Annual reports: uncertainty words, trend (pp a year)	404	+0.033	< 0.001	+0.024	< 0.001
Quarterly reports: negative words, trend (pp a year)	1,211	-0.010	0.553	+0.028	0.013
Quarterly reports: uncertainty words, trend (pp a year)	1,211	-0.016	0.183	+0.009	0.145
Quarterly reports: volatility on uncertainty, with prior volatility (pp per SD)	962	+1.299	0.300	+0.067	0.951
Quarterly reports: four-session return on negative words (pp per SD)	1,029	-2.027	0.024	-0.410	0.622

Believed: the whole-filing measures record how much risk disclosure a company prints as much as how it writes. Item 1A's share of the annual report grows +0.43 pp a year, and removing the section cuts the annual-report trends to +0.041 pp and +0.024 pp, still at $p < 0.001$. In quarterly reports the section shrinks (-1.21 pp a year, $p = 0.036$) as more companies stop restating, and the negative share of the rest of the filing rises +0.028 pp a year ($p = 0.013$) while the whole-filing measure is flat. The return association on the whole filing (-2.03 pp, $p = 0.024$) disappears excluding the section (-0.41 pp, $p = 0.622$): a company that prints its risk factors prints thousands of negative words and earns a lower return that quarter than one that refers to the annual report; the design cannot separate that from whatever else marks those quarters. Uncertainty is highest at LLY, DKNQ, JOBY and lowest at IONS, IRDM, KDK; it rises fastest at WGS, LUNR, DASH and falls fastest at KDK, TER, META (Table 7).

7. Comparison with QQQ holdings

The same pipeline was run on the 94 SEC filers among the QQQ (Nasdaq-100) constituents of 9 September 2026, 2,036 filings, 22 of the companies being in both portfolios (Appendix Tables C4 to C7). QQQ annual reports average 2.07% negative and 1.81% uncertainty words and trend upward within company at +0.041 pp and +0.024 pp a year (both $p < 0.001$); no volatility or return estimate reaches 5% (smallest $p = 0.055$); of quarterly reports with a located Item 1A, 47% restate risk factors and 37% refer to the annual report. Table 9 tests each difference in one regression on the companies held by only one portfolio, whole filing and excluding Item 1A on identical filings; Appendix Figure B7 shows the company-level distributions behind the level rows.

Table 9. QQQ holdings less ARK holdings

Coefficient on a QQQ indicator (levels), on its interaction with time (trends) or with the measure (outcomes) in one regression on the companies held by only one portfolio. Level differences with calendar-quarter effects; outcome differences with the controls of Tables 5 and 6 and company effects. Filings whose Item 1A heading was located. * Company-clustered inference; the two-way clustered covariance was not positive definite.

Result	Filings	Whole filing	p	Excluding Item 1A	p
Annual reports: negative words, level (pp)	682	-0.273	0.003	-0.051	0.383
Annual reports: uncertainty words, level (pp)	682	-0.201	< 0.001	-0.008	0.786
Annual reports: negative words, trend (pp a year)	682	-0.015	0.222	-0.034	0.002*
Annual reports: uncertainty words, trend (pp a year)	682	-0.007	0.001	-0.007	0.276*
Quarterly reports: four-session return on negative words (pp per SD)	1,829	+2.906	0.007	-1.033	0.450
Quarterly reports: volatility on uncertainty, with prior volatility (pp per SD)	1,725	-2.100	0.237	-1.112	0.522
Annual reports: volatility on uncertainty, with prior volatility (pp per SD)	627	+10.043	0.002	-0.725	0.863

Believed: the difference between the portfolios is how much risk disclosure they print. QQQ annual reports carry 0.27 pp fewer negative and 0.20 pp fewer uncertainty words on the whole filing ($p = 0.003$, $p < 0.001$), and Item 1A takes 8.0 pp less of their words (30% against 35%); excluding the section the gaps are 0.051 pp ($p = 0.383$) and 0.008 pp ($p = 0.786$). ARK's annual-report trends are steeper: by 0.007 pp a year for uncertainty on the whole filing ($p = 0.001$) and by 0.034 pp a year for negative words outside Item 1A ($p = 0.002$). The whole-filing return association differs between the portfolios (+2.9 pp per standard deviation, $p = 0.007$) and not once Item 1A is removed (-1.0 pp, $p = 0.450$). In quarterly reports the volatility coefficients do not differ ($p = 0.237$, $p = 0.522$); in annual reports they differ on the whole filing (+10.0 pp per standard deviation, $p = 0.002$) and not excluding Item 1A (-0.7 pp, $p = 0.863$).

Loughran, T., and B. McDonald, 2011, When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks, *Journal of Finance* 66, 35-65. Code, executed notebook, tests and the data workbook: github.com/robynge/FRE-GY-7871A-Assignment1